

genaut — frontier report

source CSV : logs/rerun/genaut/survey_20260625_130443.csv
corpus rows: 929

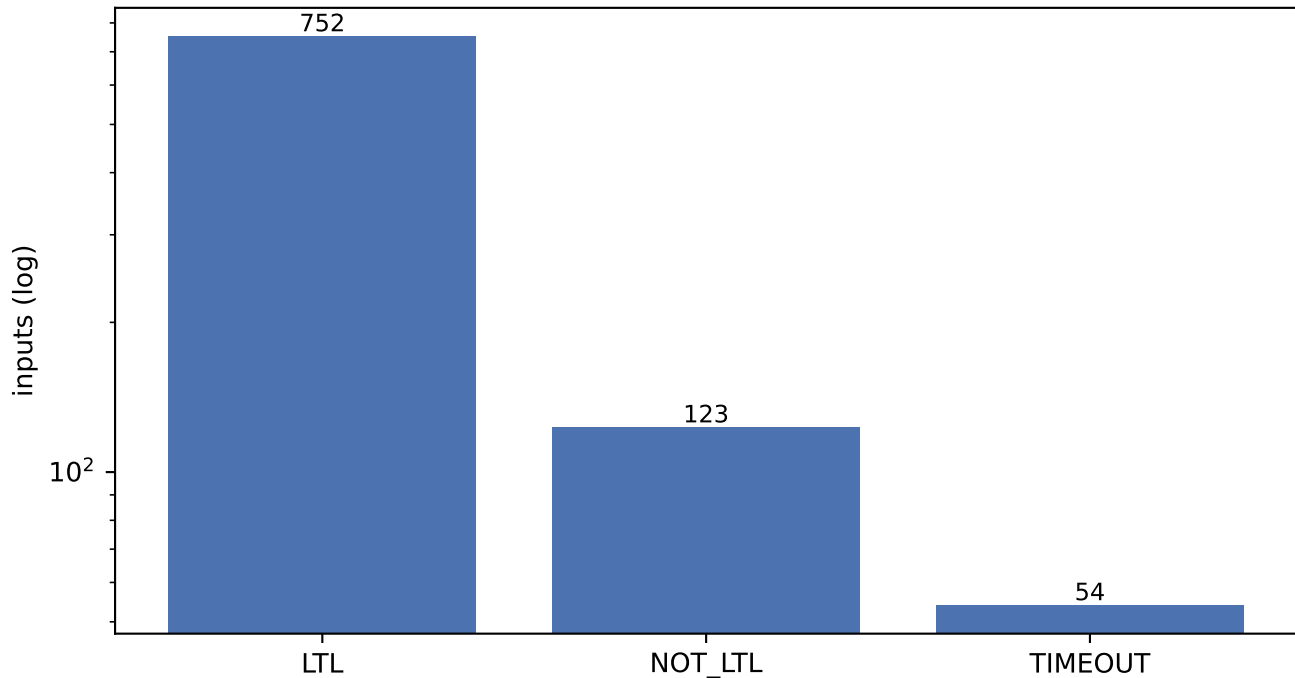
sections:

- Ventilation – LTL / not-LTL / ambiguous
- Formula collapse – structural (md5) concentration
- Size frontier – flattened formula size
- Routes to `true` – which technique built the universal 1
- Build time & timeouts
- Verification – spot-oracle validation

Ventilation — LTL / not-LTL / ambiguous

```
corpus inputs: 929
  LTL-definable (formula built) :   752  ( 80.9%)
  not-LTL (decided)             :   123  ( 13.2%)
  ambiguous (timeout, unknown)  :    54  (  5.8%)
decided (LTL or not-LTL): 875  ( 94.2%)
```

Answer frontier (by result)



Formula collapse — structural (md5) concentration

LTL answers: 752 (2 with a gated/unprintable string)

distinct structures (md5): 75

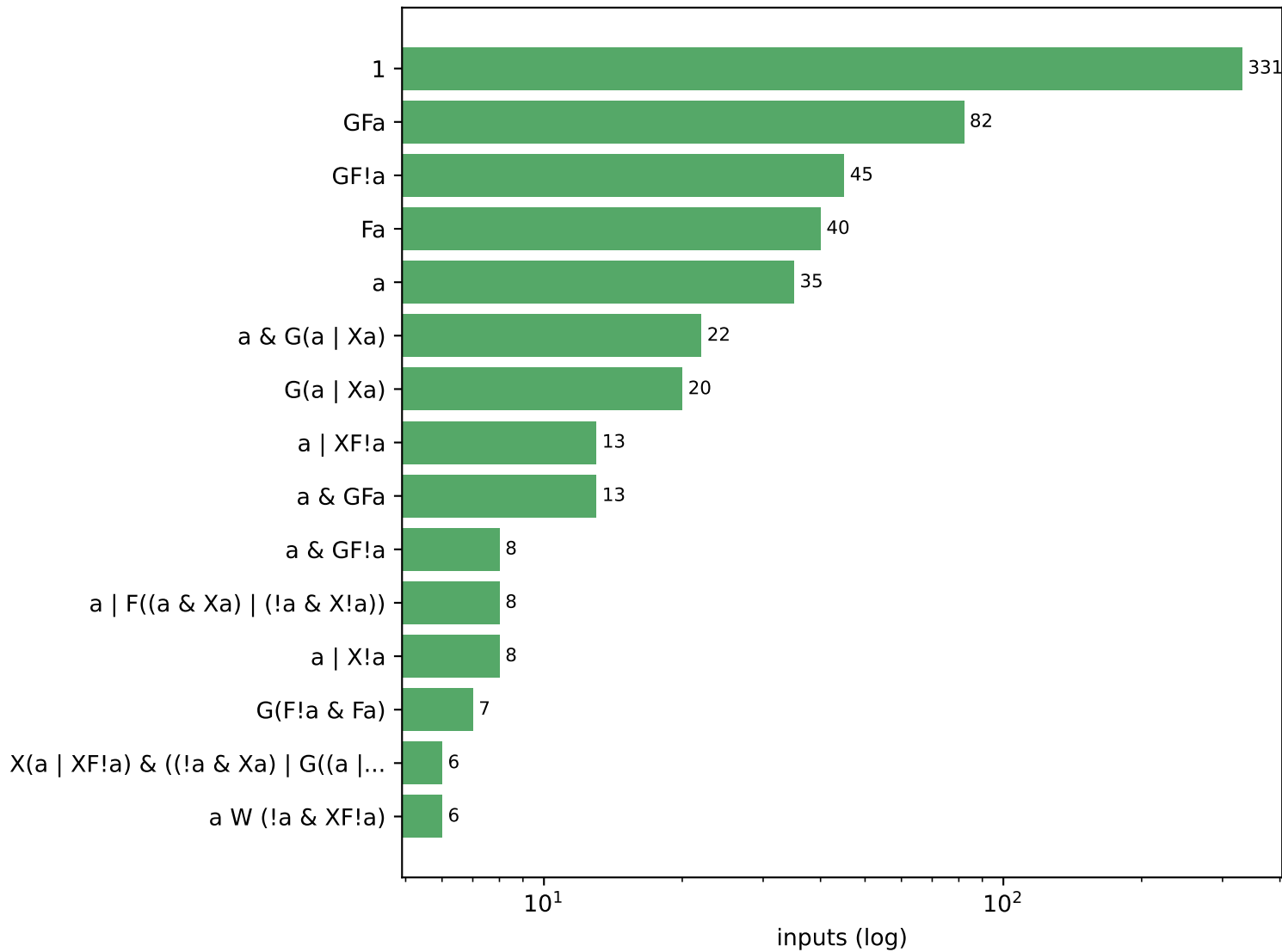
80% of answers covered by the top 10 structure(s)

90% of answers covered by the top 22 structure(s)

top 15 structures:

331	(44.0%)	1
82	(10.9%)	GFa
45	(6.0%)	GF!a
40	(5.3%)	Fa
35	(4.7%)	a
22	(2.9%)	a & G(a Xa)
20	(2.7%)	G(a Xa)
13	(1.7%)	a XF!a
13	(1.7%)	a & GFa
8	(1.1%)	a & GF!a
8	(1.1%)	a F((a & Xa) (!a & X!a))
8	(1.1%)	a X!a
7	(0.9%)	G(F!a & Fa)
6	(0.8%)	X(a XF!a) & ((!a & Xa) G((a X(XGa (!a...
6	(0.8%)	a W (!a & XF!a)

Top 15 reconstructed structures



Size frontier — flattened formula size

LTL answers with a size: 752

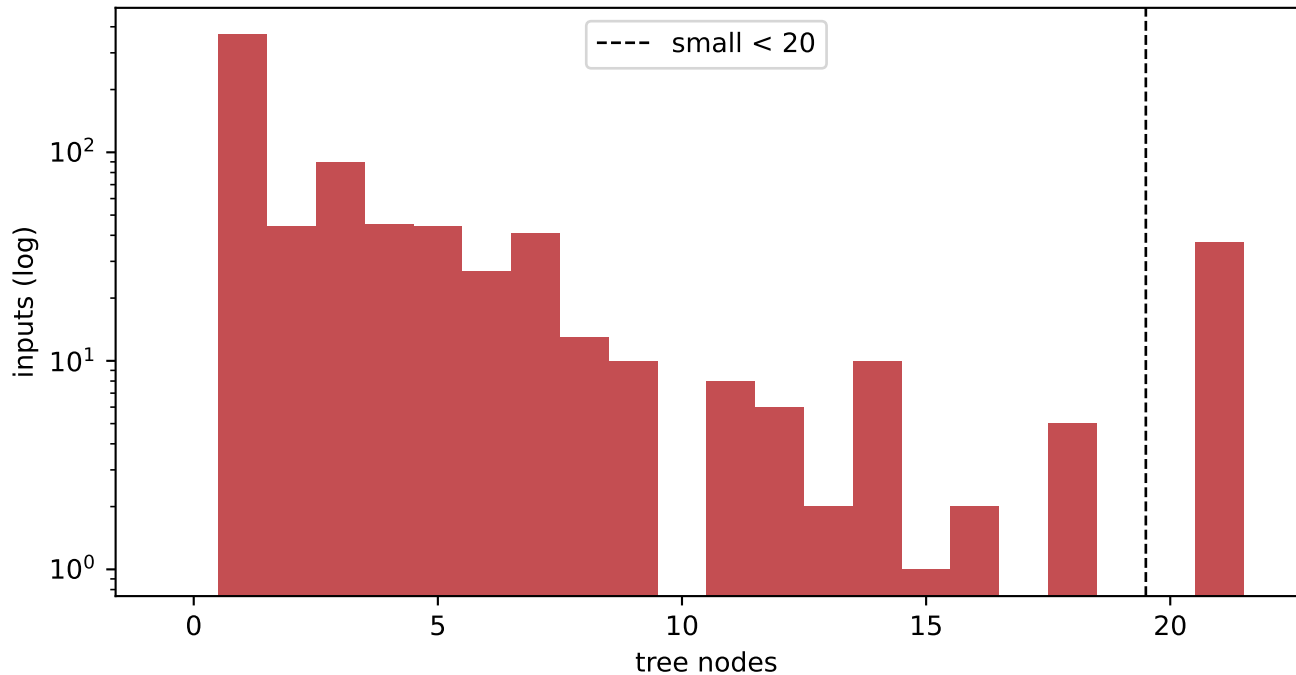
trivial (tree $\leq$ 1, i.e. true/false): 367 (48.8%)

small (tree $<$ 20): 715 (95.1%)

larger (tree $\geq$ 20): 37 (4.9%)

tree nodes: median 2 mean 747134356.4 max 370828226981

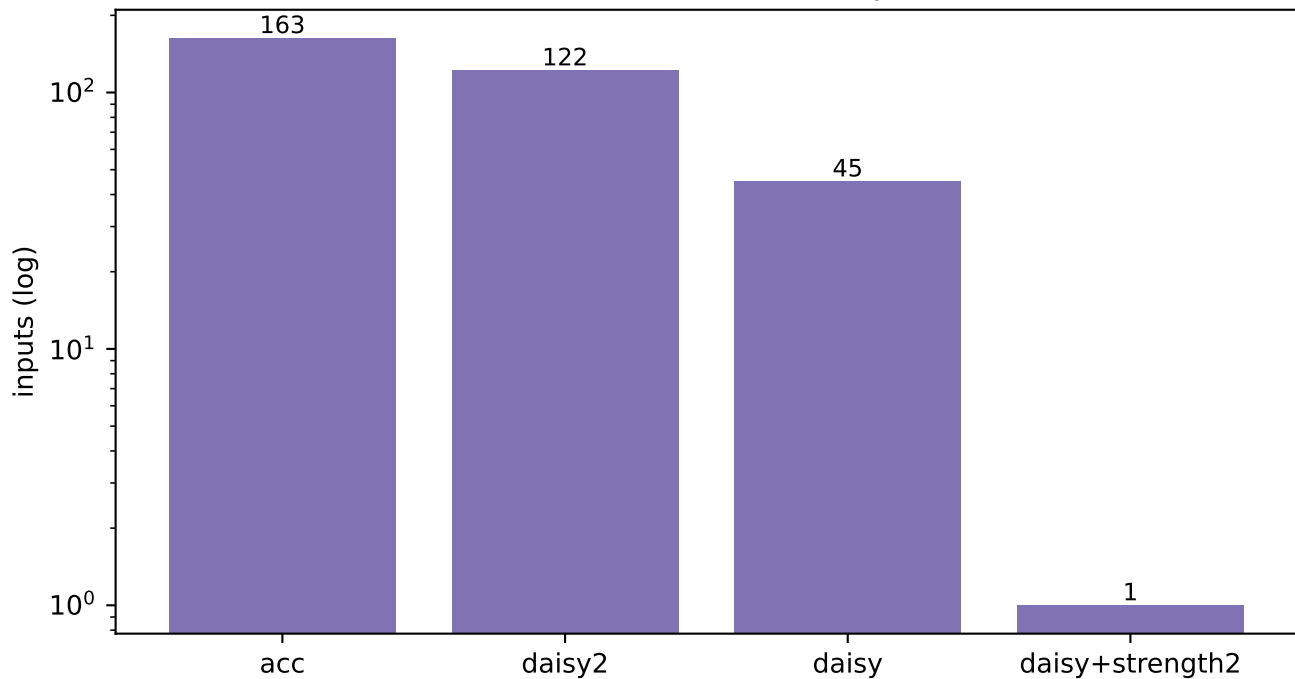
Flattened size (tree nodes); clipped at 21 (37 larger pooled at right)



Routes to `true` — which technique built the universal 1

```
universal answers (formula == '1'): 331
  163 ( 49.2%) acc
  122 ( 36.9%) daisy2
   45 ( 13.6%) daisy
    1 (  0.3%) daisy+strength2
build_s over true rows: total 209.0s  mean 0.631s  max 1.364s
```


Routes to `true` (technique)



Build time & timeouts

total build: 1569.6s mean 1.690s max 16.034s

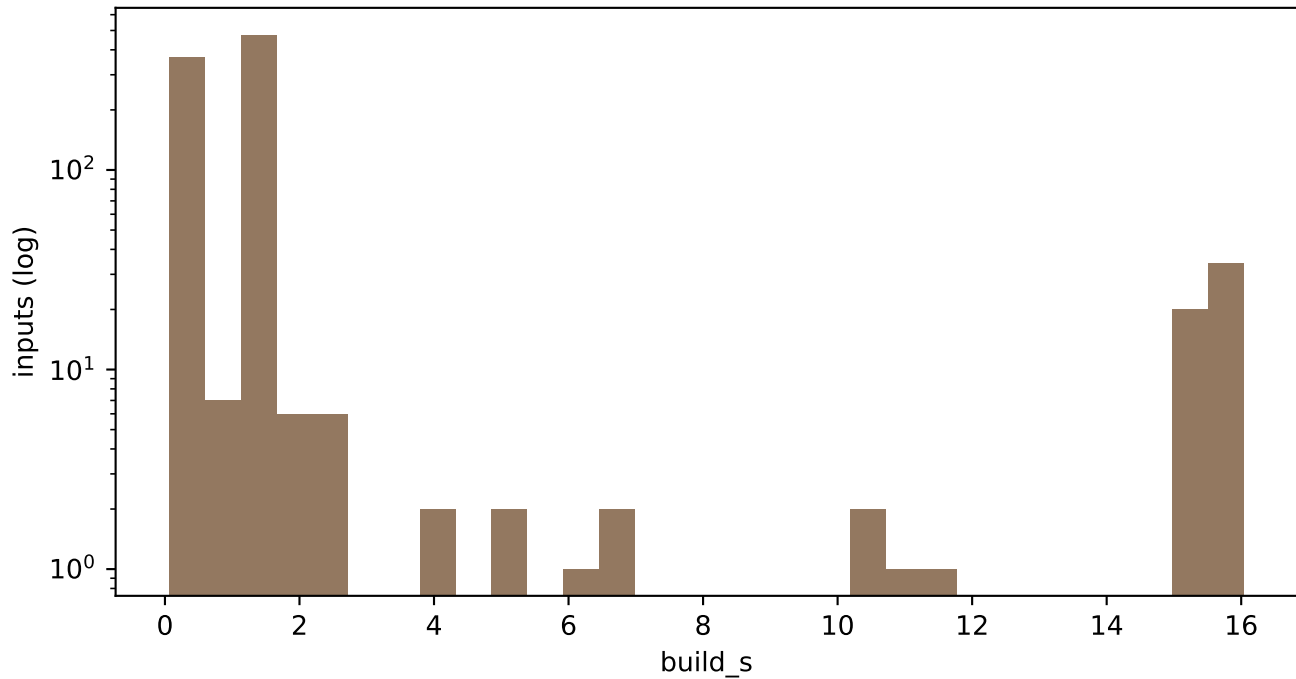
 LTL n= 752 mean 0.768s max 11.501s

 NOT_LTL n= 123 mean 1.194s max 1.259s

timeouts: 54

aut_00564.hoa
aut_00708.hoa
aut_01154.hoa
aut_01666.hoa
aut_02114.hoa
aut_02243.hoa
aut_08739.hoa
aut_08748.hoa
aut_08751.hoa
aut_08844.hoa
aut_09011.hoa
aut_09020.hoa
aut_09023.hoa
aut_09164.hoa
aut_09344.hoa
aut_09346.hoa
aut_09412.hoa
aut_10188.hoa
aut_10304.hoa
aut_10306.hoa

Build-time distribution



Verification — spot-oracle validation

validation over 752 LTL answers:
TRUE 750 (99.7%)
SIZE 2 (0.3%)
NOT-equivalent (FALSE): 0 CLEAN